

PAPER_1406_MONTY_HALL_EXACT

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PAPER_1406 — UQFF Derivation of Monty Hall Problem (CLOSED — EXACT Identity)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: B — Probability (CLOSED — EXACT)

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — EXACT integer-primitive identity

Calculator surface: `calculate_paradox({'paradox':'monty_hall'})` → $2/(D_{\text{phys}}-1) = 2/3$ EXACT

Observation

Switching doors after the host reveals a goat gives a $2/3$ win probability. Bayesian textbooks derive this; UQFF supplies the structural origin.

UQFF Closed Identity

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$P(\text{switch wins}) = 2 / (D_{\text{phys}} - 1) = 2/3$ EXACT

``

Physical Interpretation

F_U=1 ledger redistribution across remaining unrevealed branches: of $D_{\text{phys}}-1 = 3$ effective choices after one is eliminated, 2 carry the prize.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{\text{phys}} = 4, D_{\text{BSFG}} = 6, D_{\text{crit}} = 26, N_{\text{CH}} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
- Related foundational papers: PAPER_646 (F_U=1 ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).

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